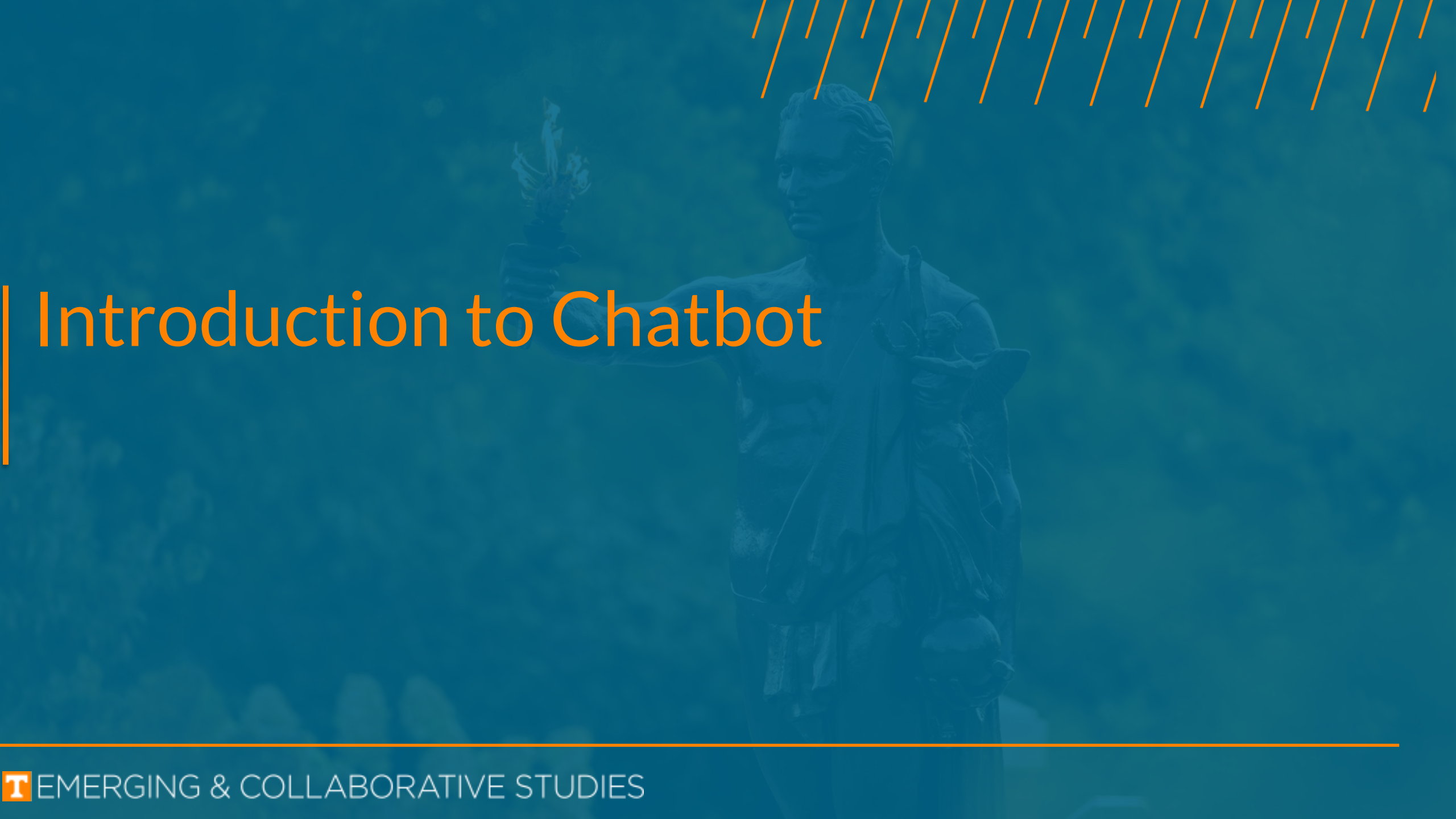

AI-102 Natural Language Processing (NLP)

Dr. Jimmy Ardiansyah



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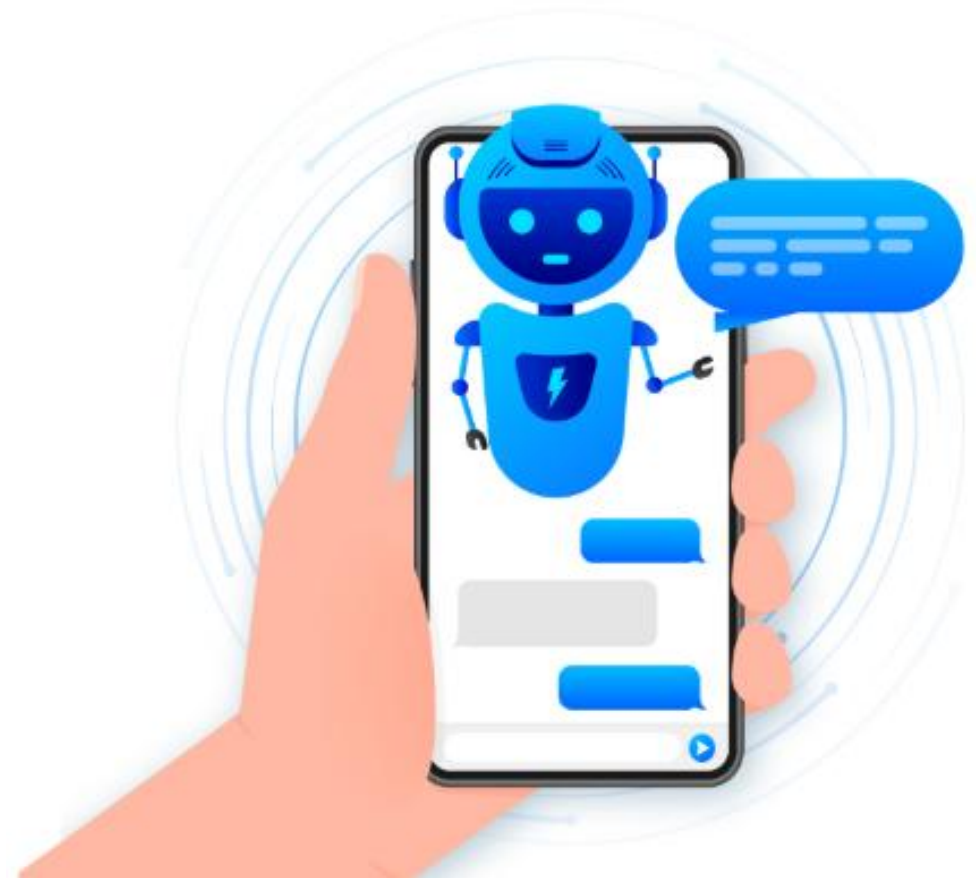




Introduction to Chatbot

What's a Chatbot

- A **chatbot** is a sophisticated software application designed to **simulate human conversation** through text or voice interactions.
- It leverages advanced **Natural Language Processing (NLP)** techniques to accurately interpret user inputs, including **context, tone, and intent**.
- Chatbots are versatile, integrating into websites, messaging apps, and mobile applications to provide **24/7 support** and automate various tasks



Types of Chatbots

Rule-Based Chatbots

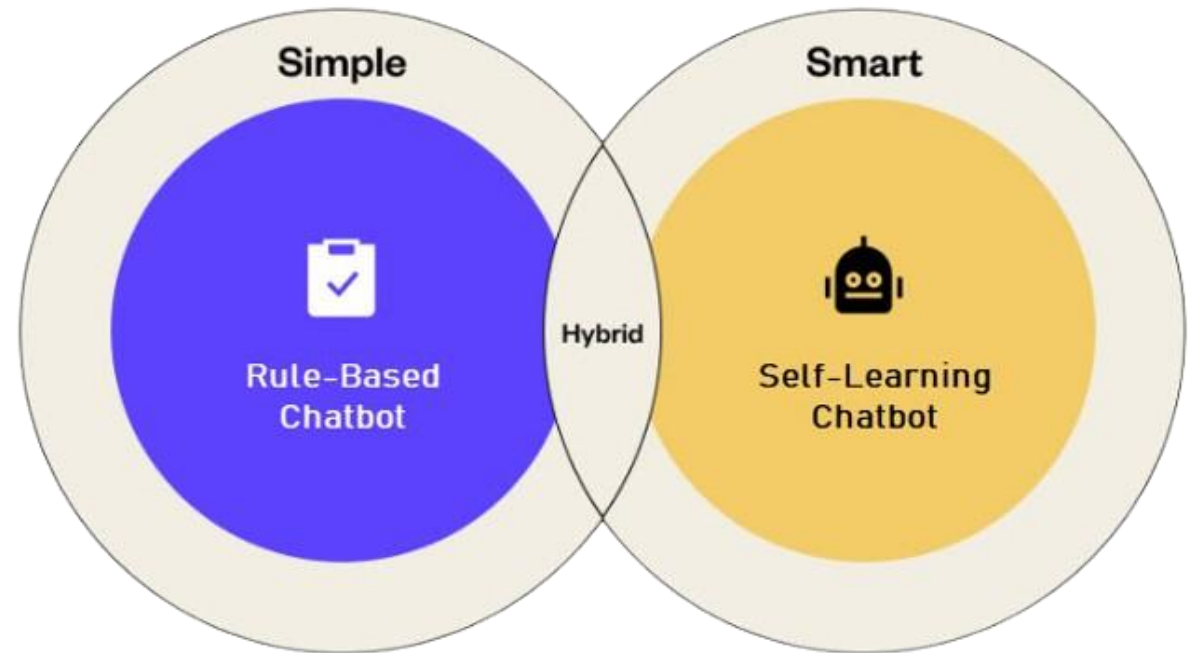
- Follow **if-then rules** / scripted flows
- Great for **simple, repetitive tasks** (FAQs, basic info)
- Easy to build, but **cannot handle complex or unseen questions**

Self-Learning (AI-Driven) Chatbots

- Use **machine learning** to understand and generate responses
- Improve over time from user interactions

Hybrid Chatbots

- Combine **rules + ML**



Where Are Chatbots Used?

- **Customer Service** – handle inquiries, solve issues, provide 24/7 support
- **E-Commerce** – product recommendations, order tracking, checkout help
- **Healthcare** – symptom checks, appointment booking, medication reminders
- **Education** – Q&A, tutoring, personalized learning paths
- **Finance** – account help, budgeting tips, alerts, basic investment info
- **Entertainment** – games, interactive stories, conversational companions



Advantages and Limitation Chatbots Models



- **24/7 Availability:** Provides round-the-clock support regardless of time zones.
- **simultaneously**, which is crucial during peak times
- **Cost-Effective:** Automates routine tasks, lowering operational costs and increasing efficiency
- **Consistency:** Delivers uniform responses, ensuring a reliable and predictable user experience



- **Limited Understanding:** Rule-based models struggle with unexpected queries, slang, or complex context
- **Lack of Empathy:** May come across as impersonal or robotic in emotional or nuanced conversations.
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